WORKED SOLUTION

STEP I 2008, Question 6

Pure • Inverse Functions and Areas

STEP I 2008, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q6

The key idea

The graph of an inverse function is the reflection of the original graph in the line $y = x$, so an integral of g with respect to y is an area measured *sideways*, between the curve $y = \mathrm{f}(x)$ and the y-axis. The upper limit k is no accident: it is exactly $\mathrm{f}(\frac{1}{2})$. Once you spot that, the two areas fit together to fill a rectangle.

Solution

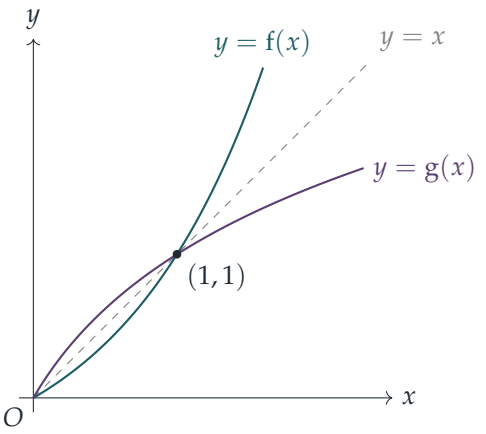
The sketch

The function f is increasing, with $\mathrm{f}(0) = 0$ and $\mathrm{f}(1) = 1$. It is convex, since $\mathrm{f}''(x) = \mathrm{e}^x/(\mathrm{e}-1) > 0$, and its gradient at the origin is $\mathrm{f}'(0) = 1/(\mathrm{e}-1) \approx 0.58 < 1$. So on $0 < x < 1$ the curve lies below the line $y = x$, it crosses that line at $(1,1)$, and then grows exponentially.

To find g, solve $y = \dfrac{\mathrm{e}^x - 1}{\mathrm{e} - 1}$ for x:

$$\mathrm{g}(x) = \ln(1 + (\mathrm{e}-1)x), \qquad x \geq 0,$$

since the range of f is $[0, \infty)$. Its graph is the reflection of $y = \mathrm{f}(x)$ in $y = x$: it passes through $(0,0)$ with gradient $\mathrm{e} - 1 \approx 1.72$, is concave, crosses $y = x$ at $(1,1)$ and then grows only logarithmically.



The first integral

$$\int_0^{1/2} \mathrm{f}(x)\,\mathrm{d}x = \frac{1}{\mathrm{e}-1}\Big[\mathrm{e}^x - x\Big]_0^{1/2} = \frac{1}{\mathrm{e}-1}\Big(\sqrt{\mathrm{e}} - \tfrac{1}{2} - 1\Big) = \frac{\sqrt{\mathrm{e}} - \frac{3}{2}}{\mathrm{e}-1}.$$

The second integral

First notice that

$$\mathrm{f}(\tfrac{1}{2}) = \frac{\sqrt{\mathrm{e}}-1}{\mathrm{e}-1} = \frac{\sqrt{\mathrm{e}}-1}{(\sqrt{\mathrm{e}}-1)(\sqrt{\mathrm{e}}+1)} = \frac{1}{\sqrt{\mathrm{e}}+1} = k,$$

using the difference of two squares $\mathrm{e}-1 = (\sqrt{\mathrm{e}})^2 - 1$. Equivalently $\mathrm{g}(k) = \frac{1}{2}$, and $1 + (\mathrm{e}-1)k = 1 + (\sqrt{\mathrm{e}}-1) = \sqrt{\mathrm{e}}$.

? **Ask yourself** How do we integrate $\ln(1 + (\mathrm{e}-1)x)$?

Substitute $u = 1 + (\mathrm{e}-1)x$, so $\mathrm{d}x = \mathrm{d}u/(\mathrm{e}-1)$, with limits $u = 1$ and $u = \sqrt{\mathrm{e}}$. Then use $\int \ln u\,\mathrm{d}u = u\ln u - u$ (integration by parts with $\ln u$ times 1):

$$\begin{aligned}\int_0^k \mathrm{g}(x)\,\mathrm{d}x &= \frac{1}{\mathrm{e}-1}\int_1^{\sqrt{\mathrm{e}}} \ln u\,\mathrm{d}u = \frac{1}{\mathrm{e}-1}\Big[u\ln u - u\Big]_1^{\sqrt{\mathrm{e}}} \\ &= \frac{1}{\mathrm{e}-1}\Big(\tfrac{1}{2}\sqrt{\mathrm{e}} - \sqrt{\mathrm{e}} - (0-1)\Big) = \frac{1 - \frac{1}{2}\sqrt{\mathrm{e}}}{\mathrm{e}-1},\end{aligned}$$

using $\ln\sqrt{\mathrm{e}} = \frac{1}{2}$.

Adding

$$\int_0^{1/2} f(x)\,\mathrm{d}x + \int_0^k \mathrm{g}(x)\,\mathrm{d}x = \frac{\sqrt{\mathrm{e}} - \frac{3}{2} + 1 - \frac{1}{2}\sqrt{\mathrm{e}}}{\mathrm{e}-1} = \frac{\frac{1}{2}(\sqrt{\mathrm{e}}-1)}{(\sqrt{\mathrm{e}}-1)(\sqrt{\mathrm{e}}+1)} = \frac{1}{2(\sqrt{\mathrm{e}}+1)},$$

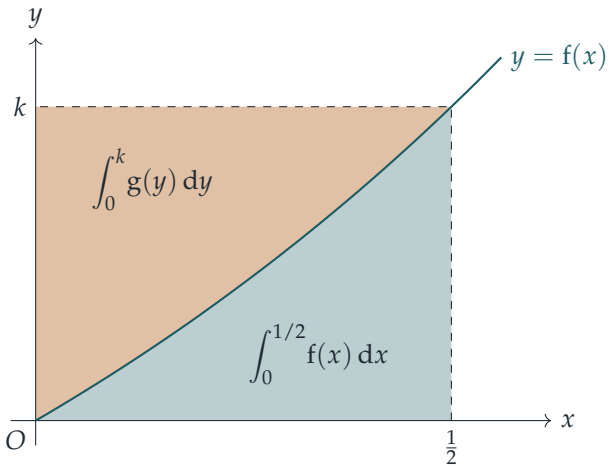
as required.

Explaining the result

The answer is $\frac{1}{2} \times k$, which is the area of the rectangle with corners $(0,0)$ and $(\frac{1}{2},k)$. Here is why.

? **Ask yourself** Where is $\int_0^k \mathrm{g}(x)\,\mathrm{d}x$ on the graph of f?

Reflecting in $y = x$ preserves area. The region under $y = \mathrm{g}(x)$ for $0 \le x \le k$ reflects to the region between the curve $y = \mathrm{f}(x)$ and the y-axis, for $0 \le y \le k$. In other words, it is $\int_0^k x\,\mathrm{d}y$ along the curve $y = \mathrm{f}(x)$: the area to the *left* of the curve. Since $\mathrm{f}(\frac{1}{2}) = k$, the curve runs from the corner $(0,0)$ to the corner $(\frac{1}{2},k)$ of the rectangle and splits it into two pieces: the area below the curve (the first integral, teal) and the area to its left (the second integral, amber).



So the two integrals together make up the whole rectangle, of area $\frac{1}{2}k = \dfrac{1}{2(\sqrt{\mathrm{e}}+1)}$. The same picture gives the general fact that, for an increasing function with $\mathrm{f}(0) = 0$,

$$\int_0^a \mathrm{f}(x)\,\mathrm{d}x + \int_0^{\mathrm{f}(a)} \mathrm{f}^{-1}(y)\,\mathrm{d}y = a\,\mathrm{f}(a).$$

Checks

Numerically, the first integral is about 0.08655 and the second about 0.10222; Simpson's rule on each agrees with the exact forms to twelve decimal places, and their sum 0.18877 equals $1/(2(\sqrt{\mathrm{e}}+1))$. As a sanity check on the sketch, both pieces are positive and the teal piece is smaller, as it should be: the convex curve sags below the diagonal of the rectangle.

Common slips

- Not noticing that $k = \mathrm{f}(\frac{1}{2})$, and so missing the simplification $1 + (\mathrm{e} - 1)k = \sqrt{\mathrm{e}}$.
- Drawing f above $y = x$ near the origin: its gradient there is $1/(\mathrm{e} - 1) < 1$.
- In the diagram, shading the area under $y = \mathrm{g}(x)$ on a separate picture without saying how it relates to the region beside $y = \mathrm{f}(x)$; the point is that reflection carries one to the other.

Beyond STEP

The identity $\int_0^a \mathrm{f} + \int_0^{\mathrm{f}(a)} \mathrm{f}^{-1} = a\,\mathrm{f}(a)$ is the equality case of *Young's inequality*: for any $b \geq 0$, $\int_0^a \mathrm{f} + \int_0^b \mathrm{f}^{-1} \geq ab$. Taking $\mathrm{f}(x) = x^{p-1}$ gives $ab \leq a^p/p + b^q/q$ with $1/p + 1/q = 1$, the starting point for Hölder's inequality in analysis. Abbott's *Understanding Analysis* is a good place to see how integrals like these are made rigorous.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.